

Hall Effect

Timothy Lim

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Introduction and Theory

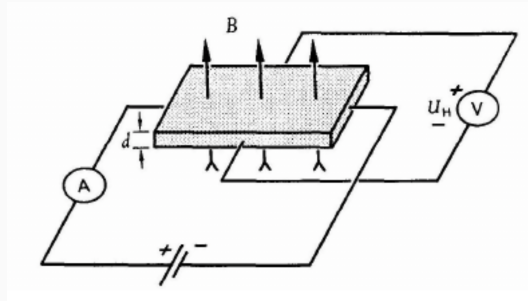


Figure 1: Hall Effect for an N-Type Semiconductor

$$\vec{F} = e(\vec{v} \times \vec{B}) \quad (1)$$

Experimental Setup

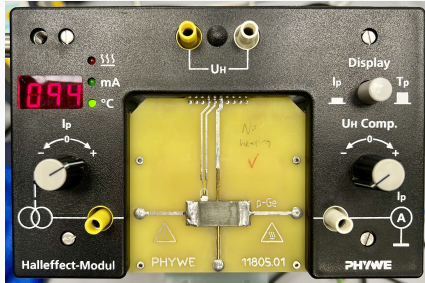


Figure 2: N-Type Semiconductor in Setup

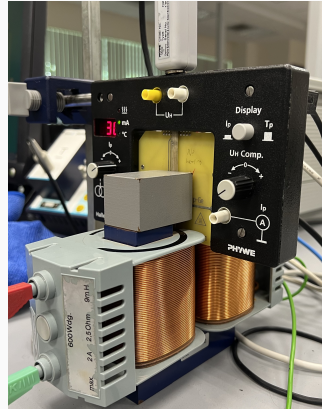


Figure 3: Full Setup with Magnetic Field and Hall Probe

Experiment 1 (Vary Current, Measure Hall Voltage)

$$\text{Hall Coefficient } H = \frac{V_H d}{BI} \quad (2)$$

$$V_H = \frac{HB}{d} I \quad (3)$$

Experiment 1 (Vary Current, Measure Hall Voltage)

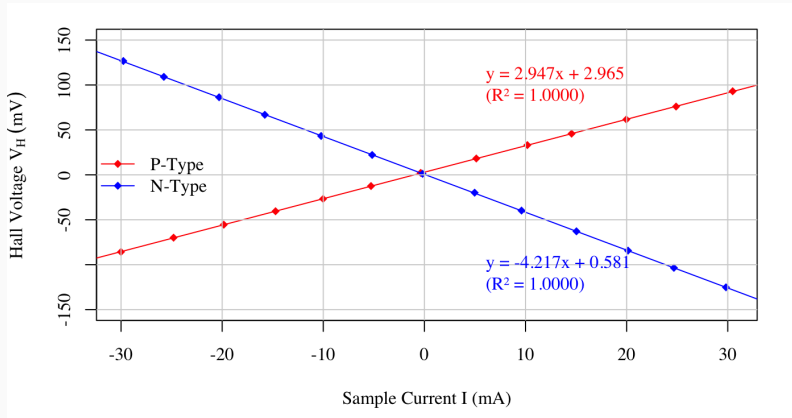


Figure 4: Plot of Hall Voltage against Sample Current for P and N-Type Germanium

Experiment 2a (Magnetoresistance)

$$R = R_0 + R_0(\mu B)^2 \quad (4)$$

$$V_s = IR_0 + IR_0\mu^2 B^2 \quad (5)$$

Experiment 2a (Magnetoresistance)

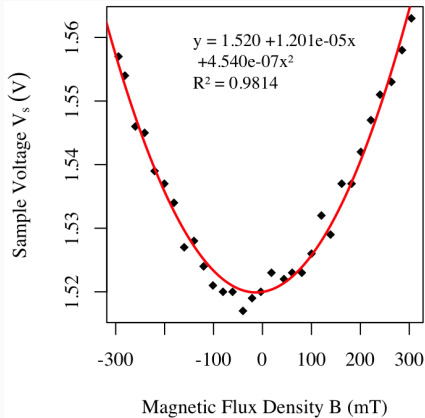


Figure 5: Sample Voltage against Magnetic Flux Density (P-Type)

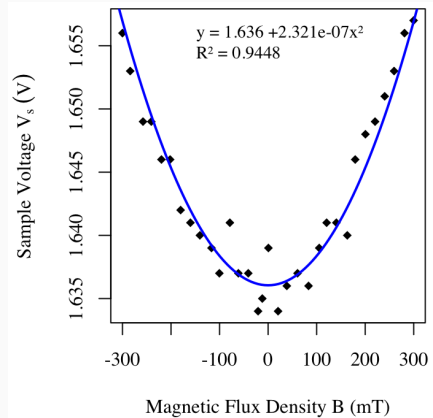


Figure 6: Sample Voltage against Magnetic Flux Density (N-Type)

Experiment 2a (Magnetoresistance)

$$R = R_0 + R_0(\mu B)^2 \quad (6)$$

$$V = IR_0 + IR_0\mu^2 B^2 \quad (7)$$

$$a = IR_0\mu^2 \quad (8)$$

$$\mu = \sqrt{\frac{a}{IR_0}} \quad (9)$$

$$\mu_{\text{Mag,P-type}} = (0.5401 \pm 0.0018) \text{ T}^{-1}$$

$$\mu_{\text{Mag,N-type}} = (0.3862 \pm 0.0014) \text{ T}^{-1}$$

Experiment 2b (Hall Effect)

$$\mu = H\sigma_0 \quad (10)$$

$$= \frac{HL}{R_0 A} \quad (11)$$

$$H = \frac{V_H d}{BI} \quad (12)$$

$$V_H = \frac{HI}{d} B \quad (13)$$

Experiment 2b (Hall Effect)

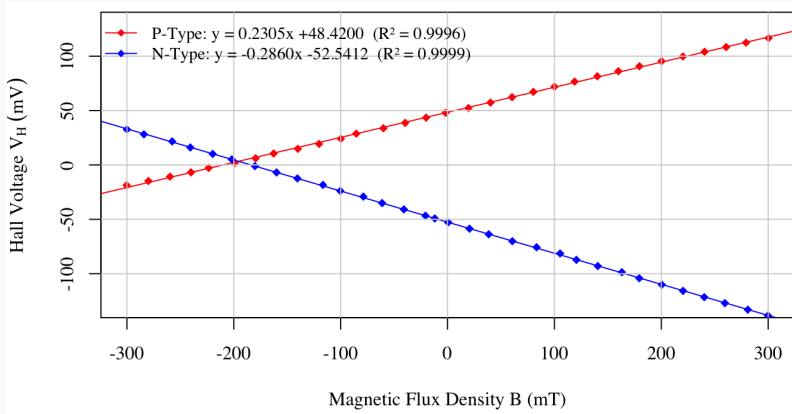


Figure 7: Plot of Hall Voltage against Magnetic Flux Density

Experiment 2b (Calculation of H)

$$H = \frac{V_H d}{BI} \quad (14)$$

$$V_H = \frac{HI}{d} B \quad (15)$$

$$H = \frac{\text{gradient} \times d}{I} \quad (16)$$

$$H_{\text{P-type}} = (7.84 \pm 0.15) \times 10^{-3} \frac{\text{m}^3}{\text{As}}$$

$$H_{\text{N-type}} = (-9.569 \pm 0.07) \times 10^{-3} \frac{\text{m}^3}{\text{As}}$$

Experiment 2b (Calculation of μ)

$$\mu = H\sigma_0 \quad (17)$$

$$= \frac{HL}{R_0A} \quad (18)$$

Where $L = 20\text{mm}$, $A = 10\text{mm}^2$

$$\mu_{\text{Hall,P-type}} = (0.310 \pm 0.006) \text{ T}^{-1}$$

$$\mu_{\text{Hall,N-type}} = (0.350 \pm 0.007) \text{ T}^{-1}$$

Experiment 2b (Calculation of n)

$$n = \frac{1}{|eH|} \quad (19)$$

$$n_{\text{P-type}} = (7.97 \pm 0.11) \times 10^{20} \text{ m}^{-3}$$

$$n_{\text{N-type}} = (6.53 \pm 0.06) \times 10^{20} \text{ m}^{-3}$$

Experiment 2b

$$\begin{aligned}\frac{\Delta H}{H} &= \sqrt{\left(\frac{\Delta H}{H}\right)_{\text{stat}}^2 + \left(\frac{\Delta I}{I}\right)^2} \quad (20) \\ &\approx \sqrt{(0.1\%)^2 + (0.67\%)^2} \\ &\approx 0.68\%\end{aligned}$$

$$\begin{aligned}\frac{\Delta \mu}{\mu} &= \sqrt{\left(\frac{\Delta H}{H}\right)^2 + \left(\frac{\Delta R_0}{R_0}\right)^2} \quad (21) \\ &\approx \sqrt{(0.68\%)^2 + (0.076\%)^2} \\ &\approx 0.68\%\end{aligned}$$

$$\frac{\Delta n}{n} = \frac{\Delta H}{H} \quad (22)$$

Experiment 3 (Constant I & B)

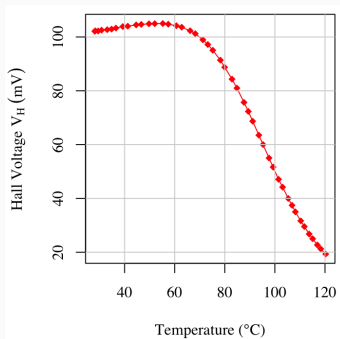


Figure 8: Hall Voltage against Temperature (P-type)

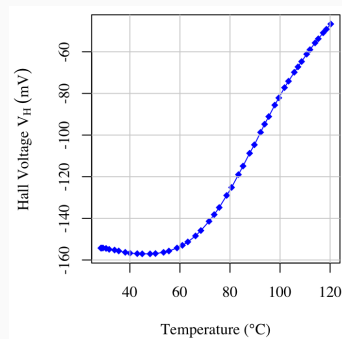


Figure 9: Hall Voltage against Temperature (N-type)

$$I = nAvq$$

$$\vec{F} = q(\vec{v} \times \vec{B})$$

Magnetic Mobility

$$\mu_{\text{Mag,P-type}} = (0.5401 \pm 0.0018) \text{ T}^{-1}$$

$$\mu_{\text{Mag,N-type}} = (0.3862 \pm 0.0014) \text{ T}^{-1}$$

Magnetic field increases path length of charge carrier, leading to more scattering and higher resistance.

Hall Mobility

$$\mu_{\text{Hall,P-type}} = (0.310 \pm 0.006) \text{ T}^{-1}$$

$$\mu_{\text{Hall,N-type}} = (0.350 \pm 0.007) \text{ T}^{-1}$$

Higher carrier mobility, higher speed, larger magnetic force, larger Hall voltage needed to balance magnetic force.

Discussion (Horizontal Offset in 2a)

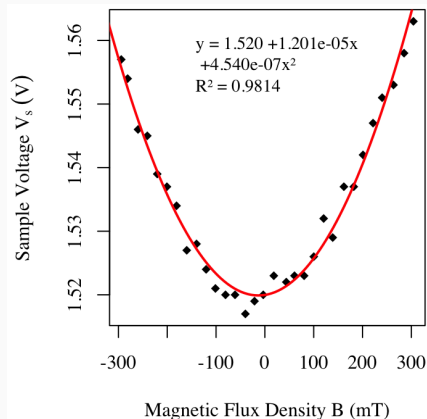


Figure 10: Sample Voltage against Magnetic Flux Density (P-Type)

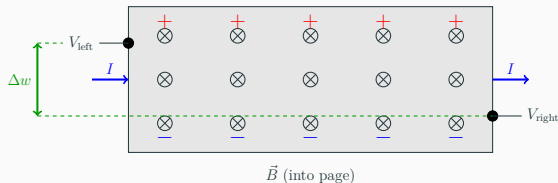
Discussion (Horizontal Offset in 2a)

$$V_s = V_0 + aB + V_0\mu^2 B^2$$

$$aB = V_{\text{hall,offset}}$$

$$\begin{aligned}\frac{\Delta w}{w} &= \frac{V_{\text{hall,offset}}}{V_{\text{hall,total}}} \\ &= \frac{aB}{\frac{HI}{d}B} \\ &= \frac{a}{m_{\text{hall}}}\end{aligned}$$

$$\begin{aligned}\Delta w &= \frac{a \cdot w}{m_{\text{hall}}} \\ &= \frac{1.201 \times 10^{-5} \cdot 1 \times 10^{-2}}{0.2305 \times 10^{-3}} \\ &= 0.521 \text{ mm}\end{aligned}$$



Discussion (Vertical Offset in 2b)

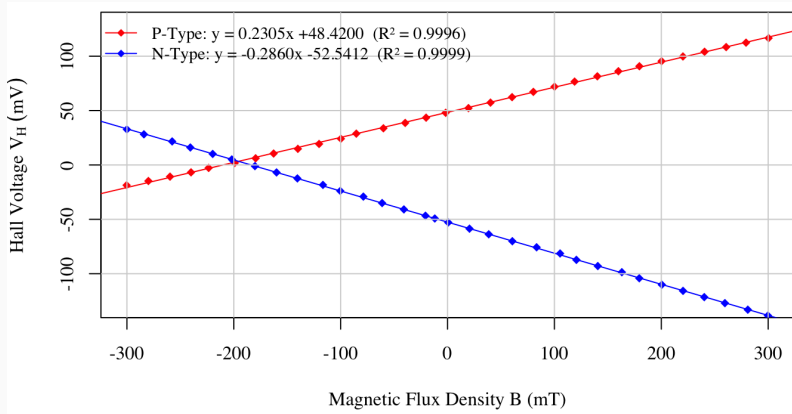


Figure 11: Plot of Hall Voltage against Magnetic Flux Density

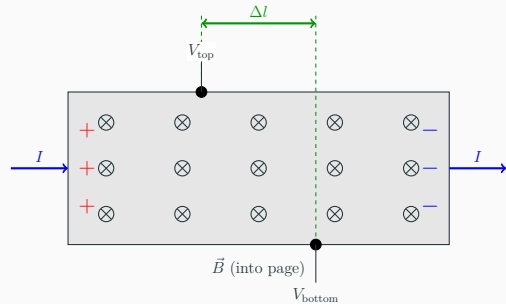
Discussion (Vertical Offset in 2b)

P-Type Sample:

$$\Delta l = \frac{V_{\text{offset}}}{V_s} \cdot l = \frac{48.4\text{mV} \times 20\text{mm}}{1.520\text{V}} = 0.637\text{mm}$$

N-Type Sample:

$$\Delta l = \frac{V_{\text{offset}}}{V_s} \cdot l = \frac{-52.5\text{mV} \times 20\text{mm}}{1.636\text{V}} = -0.642\text{mm}$$



Thank you!

Appendix: Why E is $\approx$ constant for Exp 2b

$$\begin{aligned}V_{\text{hall, measured}} &= V_{\text{hall,true}} + V_{\text{sample,leakage}} \\&= \frac{HI}{d}B + \frac{\Delta l}{l}V_s \\&= \frac{HI}{d}B + \frac{\Delta l}{l}(V_0 + V_0\mu^2 B^2) \\&= \frac{HI}{d}B + \frac{\Delta l}{l}V_0 + \frac{\Delta l}{l}V_0\mu^2 B^2 \\&= \frac{HI}{d}B + 50\text{mV} + \frac{0.6}{20} \cdot 1.6 \cdot 0.4^2 \cdot 0.3^2 \\&= \frac{HI}{d}B + 50\text{mV} + 0.69\text{mV} \\&\approx \frac{HI}{d}B + \frac{\Delta l}{l}V_0\end{aligned}$$